Tutorial 1 (Sets)

1) If A = $\{x / 6x^2 + x - 15 = 0\}$
B = $\{x / 2x^2 - 5x - 3 = 0\}$
C = $\{x / 2x^2 - x - 3 = 0\}$, then find
i. $A \cup B \cup C$
ii. $A \cap B \cap C$
Ans:
i. $A \cup B \cup C = \{-5/3,-1,-1/2,3/2,3\}$
ii. $A \cap B \cap C = \{ \}$

2) If A = {1, 2, 3, 4}, B = {3, 4, 5, 6}, C = {4, 5, 6, 7, 8} and universal set X = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, then verify the following:
i. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
ii. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
iii. $(A \cup B)' = A' \cap B'$
iv. $(A \cap B)' = A' \cup B'$
v. $A = (A \cap B) \cup (A \cap B')$
vi. $B = (A \cap B) \cup (A' \cap B)$
vii. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

3) If A, B, C are the sets of the letters in the words 'college', 'marriage' and 'luggage' respectively, then verify that $[A - (B \cup C)] = [(A - B) \cap (A - C)]$.

4) If A and B are subsets of the universal set X and n(X) = 50, n(A) = 35, n(B) = 20, $n(A' \cap B') = 5$, find
i. $n(A \cup B)$
ii. $n(A \cap B)$
iii. $n(A' \cap B)$
iv. $n(A \cap B')$.
Ans: 45, 10, 10 & 25.

5) Given that
A = {x : x is a even natural number less than or equal to 10}
and B = {x : x is an odd natural number less than or equal to 10}
Find (i) A-B (ii) B-C (iii) is A-B=B-A ?
Ans:
(i) A-B ={2, 4, 6, 8, 10}(ii) B-A ={1, 3, 5, 7, 9}

6) In a class of 200 students who appeared certain examinations, 35 students failed in MHT-CET, 40 in AIEEE and 40 in IIT entrance, 20 failed in MHT-CET and

AIEEE, 17 in AIEEE and IIT entrance,15 in MHT-CET and IIT entrance and 5 failed in all three examinations. Find how many students

i. did not fail in any examination.

ii. failed in AIEEE or IIT entrance.

Ans: 132 & 63.

7) From amongst 2000 literate individuals of a town, 70% read Marathi newspapers, 50% read English newspapers and 32.5% read both Marathi and English newspapers. Find the number of individuals who read,

i. at least one of the newspapers.

ii. neither Marathi nor English newspaper.

iii. only one of the newspapers.

Ans: 1750, 250 & 1100.

8) Out of forty students, 14 are taking English Composition and 29 are taking Chemistry.

If five students are in both classes, a) how many students are in neither class?

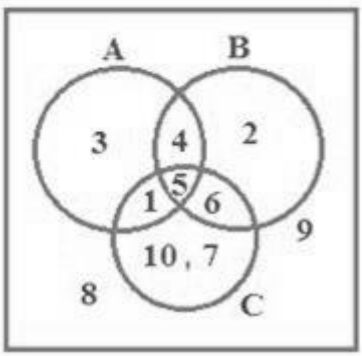
b) How many are in either class?

Ans:

a.Two students are taking neither class.

b. There are 38 students in at least one of the classes.

9) From the adjoining Venn diagram, find the following sets.



(i) A(ii) B (iii) ξ (iv) A'(v) B'(vi) C'(vii) C - A (viii) B - C (ix) A - B(x) A ∪ B (xi) B ∪ C(xii) A ∩ C(xiii) B ∩ C(xiv) (B ∪ C)'(xv) (A ∩ B)'(xvi) (A ∪ B) ∩ C (xvii) A ∩ (B ∩ C)

Ans:

(i) A= {1, 3, 4, 5}

(ii) B= {4, 5, 6, 2}

(iii) ξ= {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

(iv) A'= {2, 6, 7, 8, 9, 10}

(v) B' = {1, 3, 7, 8, 9, 10}
(vi) C' = To find C = {1, 5, 6, 7, 10}
Therefore, C' = {2, 3, 4, 8, 9}
(vii) C – A Here C = {1, 5, 6, 7, 10} A = {1, 3, 4, 5}
then C – A = {6, 7, 10}
(viii) B - C Here B = {4, 5, 6, 2} C = {1, 5, 6, 7, 10}
B - C = {4, 2}
(ix) B – A Here B = {4, 5, 2} A = {1, 3, 4, 5}
B - A = {6, 2}
(x) A ∪ B Here A = {1, 3, 4, 5} B = (4, 5, 6, 2}
A ∪ B = {1, 2, 3, 4, 5, 6}
(xi) B ∪ C Here B = {4, 5, 6, 2} C = {1, 5, 6, 7, 10}
B ∪ C = {1, 2, 4, 5, 6, 7, 10}
(xii) (B ∪ C)' Since, B ∪ C = {1, 2, 4, 5, 6, 7, 10}
Therefore, (B ∪ C)' = {3, 8, 9}
(xiii) (A ∩ B)' A = {1, 3, 4, 5} B = {4, 5, 6, 2}
(A ∩ B) = {4, 5}
(A ∩ B)' = {1, 2, 3, 6, 7, 8, 9, 10}
(xiv) (A ∪ B) ∩ C
A = {1, 2, 3, 4} B = {4, 5, 6, 2} C = {1, 5, 6, 7, 10}
A ∪ B= {1, 2, 3, 4, 5, 6}
(A ∪ B) ∩ C = {1, 5, 6}
(xv) A ∩ (B ∩ C)
A = {1, 3, 4, 5} B = {4, 5, 6, 2} C = {1, 5, 6, 7, 10}
B ∩ C = {5, 6} A ∩ (B ∩ C) = {5}

10) Let S = {1, 2, 3} then find power set of S?
Ans. P(S) = {{1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}, ∅}.

11) Let A = {a, b, d, e}, B = {b, c, e, f} and C = {d, e, f, g}
(i) Verify A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C)
(ii) Verify A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C)

12) There is a group of 80 persons who can drive scooter or car or both. Out of these, 35 can drive scooter and 60 can drive car. Find how many can drive both scooter and car? How many can drive scooter only? How many can drive car only? Ans:45

13) It was found that out of 45 girls, 10 joined singing but not dancing and 24 joined singing. How many joined dancing but not singing? How many joined both? Ans: 14

14) In a class of 40 students, 15 like to play cricket and football and 20 like to play cricket. How many like to play football only but not cricket? Ans: 20

15) If A {1, 3, 5}, then write all the possible subsets of A.
Ans: all possible subsets of A are { }, {1}, {3}, {5}, {1, 3}, {3, 5}, {1, 3, 5}

16) Given three sets P, Q and R such that:

P = {x: x is a natural number between 10 and 16},

Q = {y: y is a even number between 8 and 20} and

R = {7, 9, 11, 14, 18, 20}

(i) Find the difference of two sets P and Q (ii) Find Q - R

(iii) Find R – P (iv) Find Q – P

Ans.

P – Q = {11, 13, 15}, Q – R = {10, 12, 16}, R – P = {7, 9, 18, 20}

Q – P = {10, 16, 18}

17) It is known that at the university, 60 percent of the professors play tennis, 50 percent of them play bridge, 70% jog, 20% play tennis and bridge, 30% play tennis and jog and 40% play bridge and jog. If someone claimed that 20% of the professors jog and play bridge and tennis, would you believe this claim ? Why ?
Ans: $|A \cap B \cap C| = 10$, No

Tutorial 2: Relations

1. Given A = {1,2, 3,4} and B {x, y, z}. Let R be the following relation from A to B, R = {(l, y), (l, z), (3, y), (4, x), (4, z)}
(i) Determine the matrix of the relation. (iii)Find the inverse relation R^{-1} .
(ii) Draw the arrow diagram of R (iv) Determine the domain and range of R.

2. Define

(i) Relation (ii) Domain of Relation

(iii) Range of Relation (iv) Inverse Relation

and give suitable example.

3. Determine which of the following are relations from A = {a, b, c}toB = {1,2}

(a) R_1 = {(a, 1), (a, 2), (c, 2)} (b) R_2 = {(a, 2), (b, 1), (2, a)}

(c) R_3 = {(c, 1), (c, 2), (c, 3)} (d) R_4 = {(b, 1)}

(e) $R_5 = \varphi$ (f) $R_6 = A \times B$

4. Let 'R' be the relation from A = {1, 2, 3, 4} to B = {x, y, z} defined by R = {(1, y), (1, z), (3, y), (4, x), (4, z)}
(a) Determine the domain and range of R.
(b) Find the inverse relation R^{-1} of R.

5. Let S be the relation on the set N of positive integers defined by the equation $3x + 4y = 17$. Write S as a set of ordered pairs.

6. Let 'R' be the relation from X = {1, 2, 3, 4} to Y = {a, b, c, d} defined by

R = {(1, a), (1, b), (3, b), (3, d), (4, b)}

Find each of the following subsets of x :

(a) E = {x ; x R b} (b) F = {x ; x R d}

7. Let R and S be the relations from A = {1, 2, 3} to B = {a, b} defined by
R = {(1,a),(3,a),(2,b),(3,b)} , S = {(1,b), (2,b)}
Find $R \cup S$, $R \cap S$ and R^C.

8. Let R be the relation on the set N of +ve integers defined by the equation $x^2 + 2y = 100$ Write R as a set of ordered pairs. Find the domain of R Find range of R.
Find R^{-1} and describe R^{-1} by an equation.

9. Let A = {1, 2, 3, 4, 6} and let R be the relation on A defined by "x divides y" written x|y.

(i) Write R as a set of ordered pairs.
(ii) Draw its directed graph.
(iii) Find the inverse relation of R.

10. Explain the restriction of relation 'R'.

Let A = {a, b, c, d, e, f} and R = {(a, a), (a, c), (b, c), (a, e), (b, e), (c, e)} Find the restriction of R to B, where B = {a, b, c} is subset of A.

11. Define a relation on the set {a, b, c, d} that is
(i) reflexive and symmetric, but not transitive.
(ii) reflexive and transitive, but not symmetric.
(iii) transitive, reflexive and symmetric,
(iv) a symmetric and transitive.

12. Determine whether the relation R on the set A is an equivalence relation.
a. A = {a, b, c, d}, R = {(a,a,), (b,a), (b,b), (c,c), (d,d), (d,c)}
b. A = {1, 2, 3, 4, 5}, R = {(1,1),(1,2),(1,3),(2,1),(2,2),(3,1),(2,3),(3,3),(4,4),(3,2),(5,5)}
c. A = {1, 2, 3, 4}, R = {(1,1),(1,2),(2,1),(2,2),(3,1),(3,3),(1,3),(4,1),(4,4)}
d. Consider the set z of integers. Define aRb by $b = a^r$, for some positive integer r.
Show that R is a partial order on z, that is , show that R is :

(i) Reflexive (ii) Antisymmetric (iii) Transitive

e. Given a set S = (1, 2, 3, 4, 5), find the equivalence relation on S which generates the partition {(1, 2), (3), (4, 5)} Draw the graph of the relation.

13. If A = {1, 2, 3}. How many relations are possible on 'A' and how many of them are equivalence.

14. If A = {1, 2, 3, 4}. How many relation are there on A and how many of them are equivalence relation.

15. Define equivalence relation on a set. Let R be a relation on the set of integers defined by aRb iff a-b is a multiple of 5. Prove that R is equivalence relation.

16. Let R be the following equivalence relation on the set A = {1,2,3,4, 5, 6 }: R = {(1,1), (1,5), (2,2), (2,3), (2, 6), (3,2),(3,3),(3,6),(4,4),(5,1),(5,5), (6,2), (6,3),(6,6)}
Find the partitions of A induced by R, i.e., find the equivalence classes of R.

17. (a) .Let $A = \{1, 2, 3, 4, 5, 6\}$ and let R be the equivalence on A defined by
$R = \{(1,1), (1,5), (2,2), (2,3), (2,6), (3,2), (3,3), (3,6), (4,4), (5,1), (5,5), (6,2), (6,3), (6,6)\}$

(i) Find the partition of A induced by R.
(ii) Find the equivalence classes of R.

(b) If $\{\{a, c, e\}, \{a, d, f\}\}$ is a partition of the set $A = \{a, b, c\ d, e, f\}$, determine the corresponding equivalence relation R.

18. Determine whether the relation R on the set A is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive.

(a) $A = Z$; $a\ R\ b$ if and only if $a \le b + 1$

(b) $A = Z^+$; $a\ R\ b$ if and only if $|a - b| \le 2$

(c) $A = Z^+$; $a\ R\ b$ if and only if $a = b^k$ for some k Z^+.

(d) $A = Z$; $a\ R\ b$ if and only if $a + b$ is even.

(e) $A = Z$; $a\ R\ b$ if and only if $|a - b| = 2$

(f) A = the set of real numbers; $a\ R\ b$ if and only if $a^2 + b^2 = 4$.

19. Let R be a relation on a set of positive integers Z defined by xRy iff $3x + 4y$ is divisible by 7. Prove that R is an equivalence relation

Tutorial 3: (Functions)

1. If $(x) = 6x^3 + 4x - 5$, find $f(1), f(-2)$

Ans: $f(1) = 5$, $f(-2) = -61$

2. For the function f (x) = 2x + 1, find the range when domain = {-3, -2, -1,0,1,2,3} .

Ans: Range = {-5, -3, -1,1,3,5,7}

3. If f (x) = x^2 , $-3 \leq x \leq 3$, find its range.
Ans: Range = $\{f(x) : 0 \leq f(x) \leq 9\}$

4. Check if the following functions are odd/even x^2+1, xsinx, x^2cosx, x^3

Ans: even, even, even & odd

5. if $f(x)=3x^4-5x^2+7$ then find f(x-1).

Ans: $3x^4-12x^3+13x^2-2x+5$

6. if f(x)=f(2x+1) such that $f(x)=x^2-3x+4$ then find x.

Ans: -1 or 2/3

7. if f(x)=f(3x-1) such that $f(x)=x^2-4x+11$ then find x.

Ans: 5/4 or ½

8. if f(x)=3x+a such that f(1)=7 then find a & f(4).

Ans: a=4, f(4)=16

9.If f (x) = $\sqrt{x+1}$ and g (x) = x^2 +2 , calculate *fog* and *gof*.

Ans: $\sqrt{x^2+3}$ & x+3

10. Evaluate f(g(3)) given that f(x) = | x - 6 | + x^2 - 1 and g(x) = 2x
Ans : 35
11. Find the composite function (f o g)(x) given that
f = {(3,6) , (5,7) , (9,0)} and g = {(2,3) , (4,5) , (6,7)}
Ans: f o g = {(2 , 6) , (4 , 7)}

12. Find the composite function (f o g)(x) given that
f = {(1,6) , (4,7) , (5,0)} and g = {(6,1) , (7,4) , (0,5)}

Ans: f o g = {(6 , 6) , (7 , 7) , (0 , 0)}

13. if $f=\frac{2x+3}{3x-2}$ such that $x\neq\frac{2}{3}$ then show that (fof)(x)=x.

14. if $f=\frac{3x+2}{4x-1}$ such that $x\neq\frac{1}{4}$ then show that (fof)(x)=x.

15.I f $f(x) = 2x + 4$ and $g(x) = 4x^2 - 2x$, what is the value of:

a) $(f + g)(2)$

b) $(f + g)(-1)$

c) $(f + g)(0)$

d) $(f + g)(x)$

e) $(f + g)(2x)$

Ans:
a) 20
b) 8
c) 4
d) $4x^2 + 4$
e) $16x^2 + 4$

16. If $f(x) = 3x - 2$ and $g(x) = \frac{1}{x}$, what is the value of:

a) $2f(x) + 4g(x)$?

b) $3f(x) - g(x)$?

c) $fg(x)$?

Ans:

a) $6x - 4 + \frac{4}{x}$.

b) $9x - 6 - \frac{1}{x}$.

c) $3 - \frac{2}{x}$.

17 Evaluate f(3) given that $f(x) = | x - 6 | + x^2 - 1$
Ans. 11

18. Find $f(x + h) - f(x)$ given that $f(x) = a x + b$
Ans : ah

19 Find $(f \circ g)(x)$ given that $f(x) = \sqrt{x}$ and $g(x) = x^2 - 2x + 1$
Ans: $| x - 1 |$

20. Is the following function even, odd, or neither?

$f(x)=12x^{11}-6x^7-5x^3$

Ans. Odd

21. If $f(x) = x^2$, $0 \le x \le 3$, find its range. Determine whether it is even, odd or neither
Ans: Range = $\{f(x) : 0 \le f(x) \le 9\}$, neither.

22. If $f : R - \left\{\frac{7}{3}\right\} \to R - \left\{\frac{4}{3}\right\}$ defined as $f(x) = \frac{4x-5}{3x-7}$. Prove that 'f ' is bijective and find the rule for f^{-1}

23. If f and g be the functions from set of integers to set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. Find fog and gof.

24. Let the functions f and g be defined by $f(x) = 2x + 1$ and $g(x) = x^2 - 2$. Find the formula defining the composition function gof.

25. If $f : R-\left\{\frac{7}{5}\right\} \to R-\left\{\frac{2}{5}\right\}$ defined as $f(x)=\dfrac{2x-3}{5x-7}$. Prove that 'f ' is bijective and find the rule for f^{-1}

26. Let $f : R \to R$ defined as $f(x) = 2x^3 - 7$
$g : R \to R$ defined as $g(x) = 3x^2$

$h : R \to R$ defined as $h(x) = 5x + 4$

Find the rule of defining (i) (gof) oh (ii) fo(hog).

28.f: $R \to R$ is defined as $f(x) = x^3$

g: $R \to R$ is defined as $f(x) = 4x^2 + l$
h: $R \to R$ is defined as $h(x) = 7x-l$
find the rule of defining (hog)of, go(hof).

29. Let f: $R\to R$ $f(x) = x^3$, g:$R\to R$ $g(x) = 4x^2 + 1$, h:$R \to R$ $h(x) = 7x - 2$

Find (i) go (hog) (ii) go (hof)